

Mohammad Homaiyan

Computer Engineering Undergraduate | Artificial Intelligence • Machine Learning • Deep Learning

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PROFILE

Computer Engineering undergraduate at Shiraz University of Technology focused on Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, NLP, Multimodal AI, and intelligent software systems. Experienced with Python-based development and end-to-end ML workflows spanning data preparation, experimentation, model development, evaluation, and practical deployment. Interested in research-oriented AI and applied intelligent systems.

EDUCATION

Shiraz University of Technology — B.Sc. in Computer Engineering

Currently enrolled | GPA: 3.3 / 4.0

RESEARCH INTERESTS

Artificial Intelligence; Machine Learning; Deep Learning; Computer Vision; Video Understanding; Multimodal Learning; Natural Language Processing; Generative AI; Retrieval-Augmented Generation (RAG); Semantic Search; Intelligent Systems.

PROFESSIONAL EXPERIENCE

Artificial Intelligence Engineer — esharegar

Internship | Apr 2026 – Jun 2026 | Tehran Province, Iran

AI-focused internship experience with work associated with Artificial Intelligence and Knowledge Engineering.

Software Programmer — Perfetto Gruppo

Part-time | Jul 2025 – Sep 2025 | Remote

Software development experience with emphasis on Python programming, problem solving, and implementation.

Video Editor — SUTECH Student Scientific Association of IT

Part-time | Aug 2025 – Dec 2025

Video production and editing for the Student Scientific Association of IT at Shiraz University of Technology.

Event Organizer — ICPC

Sep 2021 – Jul 2023

Participation in organizing programming-contest activities and technical events associated with ICPC.

SELECTED PROJECTS

Combat Participant Tracking & Dropout Analysis

Computer vision pipeline for participant detection and tracking in dynamic combat scenes, with analysis of participant presence and potential dropout/disengagement events.

Technologies: Python • YOLO • OpenCV • Deep Learning • Object Detection • Object Tracking

Reference: github.com/mmd607/combat-participant-analysis

Smart Traffic Analysis

Computer vision project for traffic-scene analysis using object detection and tracking for vehicle-level and traffic-flow analytics.

Technologies: Python • YOLO • OpenCV • Computer Vision • Object Tracking

Reference: github.com/mmd607/smart_traffic_analysis

Belt Tracking

Focused computer vision project for object tracking and movement analysis.

Technologies: Python • Computer Vision • Object Tracking

Reference: github.com/mmd607/belt_tracking

Multimodal Semantic Video Search & RAG

Multimodal AI system combining video understanding, speech transcription, semantic retrieval, vector search, and an LLM-powered RAG assistant with timestamp-grounded results.

Technologies: Python • PyTorch • CLIP • Whisper • ChromaDB • FastAPI • Streamlit • RAG
Reference: Portfolio documentation

UFC Fight Intelligence & Action Analysis

Research-oriented project for fighter detection and tracking, pose estimation, action recognition, and quantitative combat-video analysis.

Technologies: Python • YOLO-Pose • MediaPipe • OpenCV • Pose Estimation • Tracking
Reference: Portfolio documentation

Real-Time Anomaly Detection Dashboard

Machine-learning dashboard for anomaly identification and interactive analytical presentation.

Technologies: Python • Machine Learning • Pandas • NumPy • Scikit-learn • Streamlit
Reference: Portfolio documentation

TECHNICAL SKILLS

Programming: Python, Java, C/C++, VHDL

Machine Learning: Supervised/unsupervised learning, feature engineering, anomaly detection, model evaluation, cross-validation, hyperparameter tuning

Deep Learning: Neural networks, CNNs, RNNs, LSTM/GRU, transfer learning, embeddings, attention, transformers

Computer Vision: OpenCV, object detection, object tracking, pose estimation, video analysis, CLIP

NLP & Generative AI: Transformers, embeddings, semantic similarity, information retrieval, LLM workflows, RAG, Whisper

Frameworks & Tools: PyTorch, TensorFlow/Keras, Pandas, NumPy, Scikit-learn, FastAPI, Streamlit, ChromaDB, Hugging Face

Engineering: Git, GitHub, Docker, Linux, REST APIs, ML workflows and deployment concepts

HONORS & ACTIVITIES

- Top 5% Academic Standing
- Association Leadership Award — Student Scientific Association
- Hackathon Finalist — Anomaly Detection Dashboard
- ICPC / competitive programming event organization and technical activities

CERTIFICATIONS

Selected credentials include Microsoft learning credentials in Computer Vision, Machine Learning concepts, and Generative AI/Agents; Python and Machine Learning coursework; Database/SQL; Web technologies; and Networking. Full certification evidence and links are maintained in the portfolio and LinkedIn documentation.

ACADEMIC DIRECTION

Interested in graduate-level research at the intersection of Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Multimodal Learning, Video Understanding, NLP, Generative AI, and retrieval-based intelligent systems.